

Agentic Delegation and the Language Frontier of Software Developers

A model, GitHub evidence, and what formal verification changes

Research question. Does agentic AI only raise productivity in languages a developer already uses, or can delegation make previously infeasible languages economical to enter?

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Alexander Quispe & Kevin Xu (2026), *Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub*. arXiv:2605.25438v2.

1. The mechanism: skill frontier $\neq$ production frontier

Unaided human-skill frontier

Languages in which the developer can personally execute with language-specific skill $s_{ik,t}$.

It is sticky because language-specific capital is costly to acquire and uncertain in a new language.

Production frontier

Languages in which the developer can *profitably deliver* using any mode currently in the production menu.

Delegated code can belong to this frontier without implying unaided mastery.

$$V_{ik,t}^g = \max_{m \in \mathcal{M}_g} V_{ik,t}^m, \quad Z_{ik,t}^g = \mathbf{1} \left[V_{ik,t}^g \geq 0 \right], \quad N_{it}^g = \sum_k Z_{ik,t}^g.$$

Economic object: an extensive participation margin, not a claim that the human suddenly learns every activated language.

2. From uncertainty to surplus: CARA–Normal certainty equivalent

Latent productivity match

$$\theta_{ik} \mid \mathcal{F}_{it} \sim N\left(\mu_{ik,t}, \frac{1}{\pi_{ik,t}}\right)$$

θ_{ik} : latent developer–language productivity match;
 $\mu_{ik,t} = E[\theta_{ik} \mid \mathcal{F}_{it}]$: posterior mean;
 $\pi_{ik,t}$: belief precision, so variance is $1/\pi_{ik,t}$;
 $s_{ik,t}$: language-specific execution skill, distinct from the uncertain match θ_{ik} .

CARA–Normal payoff

$$u(Y) = -e^{-\rho Y}, \quad Y \sim N(m, v).$$

$$CE(Y) = m - \frac{\rho}{2}v$$

Variance becomes a risk penalty.

Bridge to the production menu. Each production-mode value V^m is a **net certainty-equivalent surplus**: expected payoff minus the risk penalty minus mode-specific costs.

$$V^S = \underbrace{\omega + s\mu}_{\text{expected payoff}} - \underbrace{\frac{\rho s^2}{2\pi}}_{\text{risk penalty}} - \underbrace{b}_{\text{activation cost}}.$$

3. One opportunity, three production modes

S Solo

The developer executes.

$$V^S = \omega + s\mu - \frac{\rho s^2}{2\pi} - b$$

Expected human output is offset by activation cost and risk from the uncertain developer–language match.

C Conversational AI

The developer still executes.

$$V^C = V^S + \gamma s - r_C$$

Assistance is proportional to existing language-specific skill: useful advice must still be read, adapted, and run.

D Delegation

The agent executes share λ .

$$(1 - \lambda)s\mu + \lambda az(A)$$

Human execution is partly replaced by agent execution under human specification and verification.

The comparison is not “no AI” versus “AI”. It is $\mathcal{M}_1 = \{S, C\}$ versus $\mathcal{M}_2 = \{S, C, D\}$.

4. Solo and conversational entry: augmentation needs a foothold

$$\omega \geq T^S \equiv b - s\mu + \frac{\rho s^2}{2\pi},$$

$$T^C = T^S - (\gamma s - r_C),$$

$$T^1 = \min\{T^S, T^C\} = T^S - \max\{0, \gamma s - r_C\}.$$

Paper Assumption 1

For unfamiliar languages with skill $\underline{s}$,

$$\gamma \underline{s} - r_C \leq 0.$$

For familiar languages with skill $\bar{s}$,

$$\gamma \bar{s} - r_C > 0.$$

Unfamiliar language: $\gamma \underline{s} - r_C \leq 0 \implies T^1 = T^S$. Conversational AI does not move this entry margin *under the maintained assumption*.

5. Delegated surplus and the new threshold

$$V^D = \omega + (1 - \lambda)s\mu + \lambda az(A) - \kappa(a, s) - r_D - b \\ - \frac{\rho}{2} \left[\frac{(1 - \lambda)^2 s^2}{\pi} + \sigma_D^2(a, s, A) \right].$$

$$V^D \geq 0 \iff \omega \geq T^D \equiv b - (1 - \lambda)s\mu - \lambda az(A) + \kappa(a, s) + r_D \\ + \frac{\rho}{2} \left[\frac{(1 - \lambda)^2 s^2}{\pi} + \sigma_D^2(a, s, A) \right].$$

Execution substitution

$\lambda az(A)$ replaces part of $s\mu$.

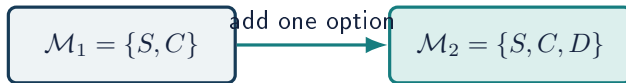
Verification cost

Delegation still requires $\kappa(a, s)$ and compute r_D .

Risk substitution

Less human-match exposure, but residual agent-error variance.

6. Proposition 1: weak frontier expansion is menu inclusion



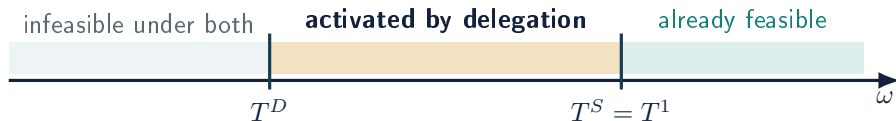
$$\mathcal{M}_1 \subset \mathcal{M}_2 \implies V_{ik,t}^2 \geq V_{ik,t}^1 \implies Z_{ik,t}^2 \geq Z_{ik,t}^1 \implies N_{it}^2 \geq N_{it}^1.$$

Why only weak? If delegation is unattractive, the developer ignores it. The result holds **path by path**, without a distributional assumption.

Strict expansion at date t additionally needs some language with $V_{ik,t}^1 < 0 \leq V_{ik,t}^D$.

7. Proposition 2: the delegation advantage creates an activation band

$$B \equiv T^S - T^D = \lambda[az(A) - s\mu] - \kappa(a, s) - r_D + \frac{\rho}{2} \left[\frac{(2\lambda - \lambda^2)s^2}{\pi} - \sigma_D^2(a, s, A) \right]$$



$$B > 0 \implies Z_{ik,t}^2 - Z_{ik,t}^1 = \mathbf{1}[T_{ik,t}^D \leq \omega_{ik,t} < T_{ik,t}^S].$$

A positive-width band is not yet strict expected expansion: the opportunity distribution must put positive mass inside it.

8. What I verified by hand

Relabeled here into the paper's printed notation.

$$\begin{aligned} B &= T^S - T^D \\ &= \left[b - s\mu + \frac{\rho s^2}{2\pi} \right] - [b - (1 - \lambda)s\mu - \lambda az(A) \\ &\quad + \kappa(a, s) + r_D + \frac{\rho}{2} \left(\frac{(1 - \lambda)^2 s^2}{\pi} + \sigma_D^2 \right)] \\ &= \lambda[az(A) - s\mu] - \kappa(a, s) - r_D + \frac{\rho}{2} \left[\frac{(2\lambda - \lambda^2)s^2}{\pi} - \sigma_D^2 \right]. \end{aligned}$$

Hand checks

- ▶ rearrange $V^D \geq 0$ without flipping a sign;
- ▶ cancel the common activation cost b ;
- ▶ use $1 - (1 - \lambda)^2 = 2\lambda - \lambda^2$;
- ▶ recover $B > 0 \Rightarrow T^D < T^S$ and the half-open band.

Scope of this evidence

The handwritten PDF is a human algebra check. It is **not** itself machine-verified; Lean separately checks formal declarations encoded from the model.

9. Mandatory Lean link: Proposition 2, objects and conclusion

Paper equation (8):

$$Z_{ik,t}^2 - Z_{ik,t}^1 = \mathbf{1}[T_{ik,t}^D \leq \omega_{ik,t} < T_{ik,t}^S], \quad B_{ik,t} > 0.$$

```
theorem threshold_indicator_sub_eq_band
  {delegationThreshold soloThreshold opportunity : ℝ}
  (hthreshold : delegationThreshold < soloThreshold) :
  ((if delegationThreshold ≤ opportunity then 1 else 0) : ℤ) -
    (if soloThreshold ≤ opportunity then 1 else 0) =
    if delegationThreshold ≤ opportunity ∧ opportunity <
      soloThreshold then 1 else 0 := by
  by_cases hdelegation : delegationThreshold ≤ opportunity
  • by_cases hsolo : soloThreshold ≤ opportunity
```

Real contiguous fragment from `lean/MainTheorems.lean`.

Economic ↔ Lean

T^D	delegationThreshold
$T^S = T^1$	soloThreshold
ω	opportunity
$B > 0$ implies	
$T^D < T^S$	hthreshold
activation	two threshold indicators

Exactness statement

The fragment proves the exact pointwise band identity conditional on $T^D < T^S$. Assumption 1 supplies $T^1 = T^S$ outside this theorem. The full Proposition-2 interface also proves the atomless-measure/CDF claim and is judged **Matches**.

10. Lean finding #1: Proposition 3 is not strictly true on its printed domain

Printed dynamic claim:

$$\Delta C_i(s) = \sum_{k \in \mathcal{U}_i} [(1 - p_{ik}^1)^{s+1} - (1 - p_{ik}^2)^{s+1}] \geq 0,$$

with strict increase and concavity in the closed-frontier benchmark $p_{ik}^1 = 0 < p_{ik}^2$.

Two missing conditions

- ▶ If $\mathcal{U}_i = \emptyset$, the sum is identically zero.
- ▶ The printed domain allows $p^2 = 1$; then the gap reaches one immediately and is constant afterward.

Corrected strict result

Lean proves strict growth and strict concavity when the frontier is nonempty and every relevant hazard satisfies

$$0 < p_{ik}^2 < 1.$$

These are **additional conditions**, not printed assumptions.

```
theorem closed_frontier_gap_strict_mono
  {Language : Type*} (unfamiliar : Finset Language) (hazard
    : Language → ℝ)
  (hne : unfamiliar.Nonempty)
  (hpos : ∀ language ∈ unfamiliar, 0 < hazard language)
  (hlt : ∀ language ∈ unfamiliar, hazard language < 1) (
    horizon : ℕ) :

theorem dynamic_strictness_boundary_counterexample :
  ¬ ((1 - (1 - (1 : ℝ)) ^ (0 + 1)) < (1 - (1 - (1 : ℝ)) ^
    (0 + 2))) := by
  norm_num
```

Verbatim excerpts from `lean/MainTheorems.lean`; the first conclusion is omitted. `closed_frontier_gap_strict_concave` uses the same conditions.

11. Lean finding #2: strict repository expansion needs opportunity mass

What the paper displays

Lower entry costs imply weakly more expected repository access; it states strict expansion when some repositories require languages in the activation band.

$$c_r(\ell(r), 2) \leq c_r(\ell(r), 1) \\ \implies \mathbb{E}[R_i^2] \geq \mathbb{E}[R_i^1].$$

What strictness additionally needs

For at least one repository,

$$\Pr\left(c_r^2 \leq \Omega_{ir,t} < c_r^1\right) > 0.$$

The source proof uses this premise; the displayed proposition omits it.

Declaration

```
theorem repository_expected_expansion_strict
```

Decisive assumption

```
(hband : ∃ repository ∈ repositories,  
  0 < (opportunityLaw repository).real  
  (Set.Ico (newCost repository) (oldCost repository)  
  )) :
```

Two verbatim excerpts from the same declaration in `lean/MainTheorems.lean`; intervening parameters and the conclusion are omitted.

Lean verdict

`repository_expected_expansion` proves the paper's weak claim as stated. The displayed fragment proves strict expansion only after adding positive mass in a lowered-cost interval.

12. Empirical evidence: sharp associations at first detectable use

+2.528

active programming languages

event time $t = 0$

+1.193

newly used languages

event time $t = 0$

+0.382

Shannon language entropy

event time $t = 0$

Design and measurement

- ▶ 5,346 developers; 28 months; 57.2 million changed files.
- ▶ Adoption: first commit with the machine-readable Co-Authored-By: Claude trailer.
- ▶ Doubly robust staggered-adoption event study, not-yet-treated comparisons, one month of anticipation.

Economic mapping

- ▶ active languages → production-set expansion;
- ▶ newly used languages → activation-band analogue;
- ▶ entropy → nontrivial diversification.

Interpretation boundary

These are **event-time associations**, not definitive causal effects. Adoption may occur when an unfamiliar-language project shock arrives.

13. Where I did not believe the AI

1. Citation / hallucination test

The initial title, *Coding Beyond Your Training*, was not fabricated: it was the genuine v1 title. I did not infer v2 authors, sample, or results until checking the source.

2. Notation correction

The v2 paper prints ω (opportunity), π (precision), ρ (risk aversion), γ (conversational gain), λ (delegated share), and $\kappa(a, s)$ (verification cost). Repository summary/hand notes re-letter several of these; this deck follows the paper.

3. “ T^D does not require skill”

Not as mathematical independence. T^D still depends on s through retained human execution, $\kappa(a, s)$, remaining human-match risk, and $\sigma_D^2(a, s, A)$. The defensible claim is narrower: the principal delegated-output term $\lambda az(A)$ does not require a positive language-specific foothold.

4. Different margin from Aouad–Lykouris–Zhong

Aouad–Lykouris–Zhong: AI substitutes for human implementation input inside an already feasible task.

Quispe–Xu: delegation can relax an entry/participation constraint and change which language–task opportunities are feasible.

14. Final takeaway: frontier expansion, with conditions

Agentic AI can expand a developer's feasible production frontier by lowering the entry threshold for unfamiliar-language opportunities.

It need not expand unaided human skill, and it need not activate opportunities with no probability mass in the lowered-threshold region.

Claim	Formal-verification takeaway
Proposition 1	Weak pathwise frontier expansion follows exactly from menu inclusion.
Proposition 2	The activation-band identity and nonnegative expected expansion are proved as stated.
Proposition 3	Strict dynamics require $\mathcal{U}_i \neq \emptyset$ and $0 < p_{ik}^2 < 1$; the printed strict clause is false at allowed boundaries.
Proposition 5	Weak repository expansion is valid; strict expansion requires positive opportunity mass in a lowered-cost interval.

EconCSLib closeout status

Partially formalized; human review 0/9. **Fast check passed**; no sorry/admit. Passing the encoded proofs does **not** establish semantic equivalence to every paper claim.